

Vidya Vikas Institute of Engineering & Technology, Mysore



VTU Question Papers

B.E.– Civil Engineering Stream

I Semester

Dec/Jan - 2024

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Vidya Vikas institute of Engineering & Technology

Bannur road, Alanahalli, Alanahalli post, Mysore- 570028

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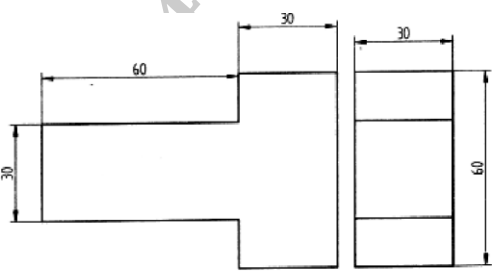
S/L	Subject Code	Subject Title	Exam Date
1	BCEDK103/203	COMPUTER AIDED ENGINEERING DRAWING	Dec.23/Jan.24
2	BCHEC102/202	Applied Chemistry for Civil Engineering Stream	Dec.23/Jan.24
3	BCIVC103/203	Engineering Mechanics	Dec.23/Jan.24
4	BENGK106/BENGK206/22BD17	Communicative English	Dec.23/Jan.24
5	BESCK104A/BESCKA104	Introduction to Civil Engineering	Dec.23/Jan.24
6	BETCK105B/BETCKB105	Green Buildings	Dec.23/Jan.24
7	BICOK107/207	Indian Constitution	Dec.23/Jan.24
8	BMATC101	Mathematics - I for Civil Engineering Stream	Dec.23/Jan.24

CBCS 2022 – SCHEME**BCEDK103/203**USN

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First/Second Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan.2024**COMPUTER AIDED ENGINEERING DRAWING****Time: 3 Hours****(COMMON TO ALL BRANCHES)****Max.Marks:100****Note:** 1. Answer all four full question.

2. Grid sheets may be provided for making preparatory sketches.

Module – 1		
Q. No.		Marks
1	A circular lamina of 30 mm diameter rests on HP such that the surface of the lamina is inclined at 30° to HP and the diameter passing through the corner on which the lamina rests on HP is inclined at 45° to VP. Draw its top and front views in this position.	20
Module - 2		
2	A pentagonal pyramid 25 mm sides of base and 50 mm axis length rests on HP on one of its corners of the base such that the two base edges containing the corner on which it rests make equal inclinations with HP. Draw the projection of the pyramid when the axis of the pyramid is inclined to HP at 40° and appears to be inclined to VP at 45° .	30
Module – 3		
3	Following figure shows the front and side views of solid. Draw the isometric projection of the solid. 	25
Module - 4		
4	A square prism of base sides 30 mm and axis length 60 mm is resting on HP with all the vertical faces equally inclined to VP. It is cut by an inclined plane 60° to HP and perpendicular to VP and is passing through a point on the axis at a distance of 50 mm from the base. Obtain the development for the truncated portion of the solid.	25

Examiner 1:

Name:

Signature:

Examiner 2:

Name:

Signature:

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First/Second Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan.2024

Applied Chemistry for Civil Engineering Stream

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
 2. VTU Formula Hand Book is permitted.
 3. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	What is Glass? Describe the preparation of Soda Lime Glass.	07	L2	CO1
	b.	Explain the various steps involved in the manufacture of cement by wet process with a flow chart.	06	L2	CO1
	c.	Define Refractories. Write the properties and applications of refractory materials.	07	L3	CO1
OR					
Q.2	a.	What is Cement? Illustrate the process of setting and hardening of cement with chemical reactions.	07	L3	CO1
	b.	Define alloys. Write the properties and applications of Iron and its alloys.	07	L3	CO1
	c.	Write a note on additives used in the manufacture of cement.	06	L3	CO1
Module – 2					
Q.3	a.	Illustrate the construction and working of Methanol – Oxygen fuel cell.	06	L4	CO2
	b.	Define corrosion. Describe the electrochemical corrosion of steel in concrete.	07	L2	CO2
	c.	What is anodizing? Explain anodizing of aluminium. Mention its applications.	07	L2	CO2
OR					
Q.4	a.	Define PV cell. Illustrate the construction and working of Photovoltaic Cell.	07	L3	CO2
	b.	Explain differential metal and aeration corrosion with suitable examples.	07	L2	CO2
	c.	Explain how material selection and design can prevent corrosion.	06	L2	CO2
Module – 3					
Q.5	a.	100 ml of a water sample required 20ml of 0.01 M EDTA for the titration with Erichrome Black-T indicator, 100 ml of the same water sample after boiling and filtering required 10 ml of 0.01 M EDTA. Calculate (i) Total hardness (ii) Permanent Hardness (iii) Temporary Hardness of the sample.	07	L3	CO3
	b.	With a neat labeled diagram illustrate the softening of hard water by ion exchange method.	07	L3	CO3
	c.	Explain the following size dependent properties of nanomaterials: (i) Catalytic property (ii) Surface area	06	L2	CO3
OR					
Q.6	a.	What is desalination? Explain desalination of brackish water by forward osmosis.	07	L2	CO3
	b.	Define Nanomaterials. Demonstrate the synthesis of Nanomaterials by Sol-gel method.	07	L3	CO3
	c.	Write a note on use of metal-oxide nano particles in the treatment of water.	06	L3	CO3

Module – 4					
Q.7	a.	Calculate the number average molecular mass (M_n) and weight average molecular mass (M_w) of a polymer in which 30% molecules have a molecular mass 20,000 ; 40% have 30,000 and the rest have 60,000.	06	L3	CO4
	b.	Define Fibers. Explain the synthesis, properties and applications of Nylon Fibers.	07	L2	CO4
	c.	Define Polymer Composites. Write the properties and applications of Fiber Reinforced Polymer (FRP) and Geo-Polymer Concrete (GPC).	07	L3	CO4
OR					
Q.8	a.	Explain the synthesis, properties and applications of Chloropolyvinyl chloride.	06	L2	CO4
	b.	Define Biodegradable Polymer. Explain the steps involved in the preparation of polylactic acid and mention the applications.	07	L2	CO4
	c.	What are adhesives? Explain the synthesis, properties and applications of epoxy resin.	07	L2	CO4
Module – 5					
Q.9	a.	State Phase Rule. Explain the terms involved in the phase rule with examples.	07	L2	CO5
	b.	With the help of a neat phase diagram, explain the Lead-Silver system.	07	L2	CO5
	c.	Describe the determination of pH of soil sample using pH sensors.	06	L2	CO5
OR					
Q.10	a.	State and explain phase rule for two component system. Mention the use of phase diagram.	06	L2	CO5
	b.	Illustrate the principle and instrumentation of conductometric sensors.	07	L2	CO5
	c.	Explain the applications of potentiometric sensors in the estimation of iron.	07	L2	CO5

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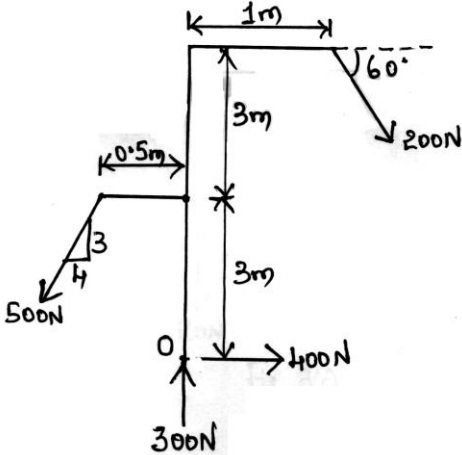
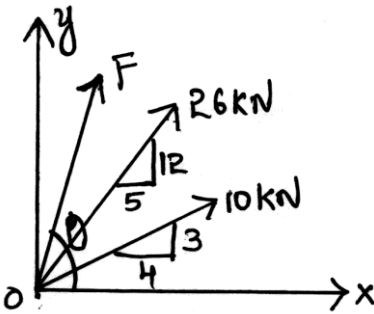
First/Second Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan.2024 Engineering Mechanics

Time: 3 hrs.

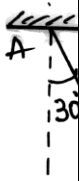
Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

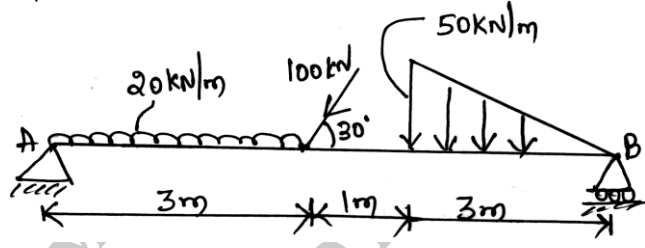
2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	What are the basic idealization of engineering mechanics, explain each	7	L2	CO1
	b.	Determine magnitude, direction and position of Resultant for the force system shown in Fig Q1(b) with respect to point 'O'.  Fig Q1(b)	10	L3	CO2
	c.	State the principle of transmissibility of forces.	3	L2	CO1
OR					
Q.2	a.	State and prove Varignon's theorem of moments.	7	L2	CO1
	b.	Define couple. What are the characteristics of couple?	3	L2	CO2
	c.	The 26kN force is the resultant of two forces, one of which is shown in Fig. 2(c). Determine other force.  Fig Q2(c)	10	L3	CO1

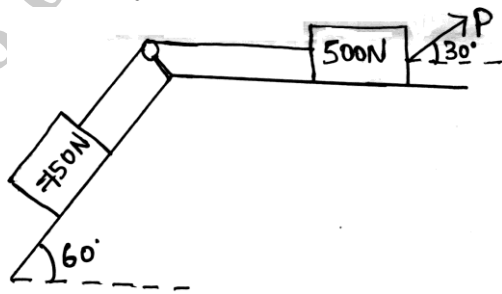
Module – 2

Q.3	a.	What are the different types of supports? Explain with the help of neat sketch.	6	L2	CO2
	b.	What are the conditions for equilibrium of concurrent force system? Also explain equilibrant force.	4	L2	CO2
	c.	Calculate tension in the strings. Also calculate angle θ in Fig 3(c) Fig Q3(c)	10	L3	CO2

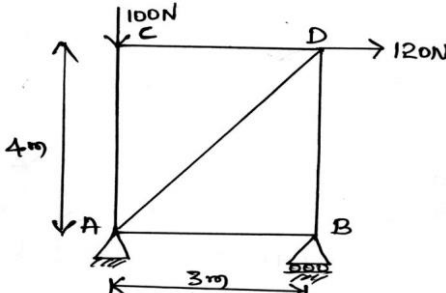
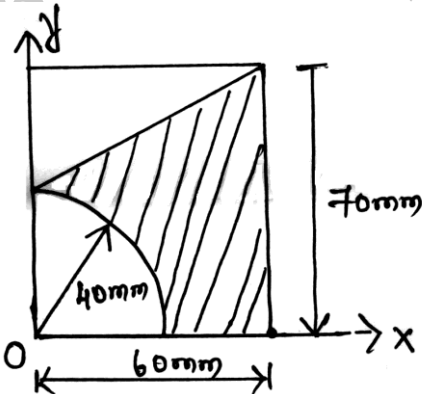
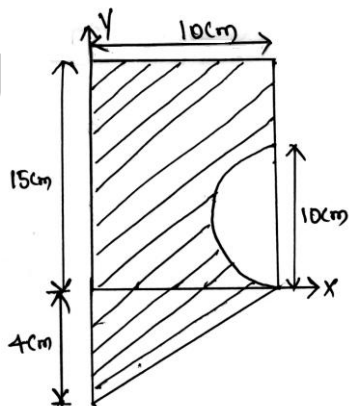
OR

Q.4	a.	State and prove Lami's theorem.	6	L2	CO2
	b.	Differentiate statically determinate and indeterminate beams.	4	L3	CO2
	c.	Determine support reaction for the beam shown in Fig Q4(c) 	10	L3	CO2

Module – 3

Q.5	a.	Describe the assumptions made in analysis of truss, by mentioning the types of trusses.	8	L2	CO3
	b.	What is the value of P in the system shown in Fig Q5(b) to cause the motion to impend? Assume the pulley is smooth and coefficient of friction between contact surfaces is 0.2. 	12	L3	CO3

OR

Q.6	a.	Give step by step procedure need to be followed in method of joints.	8	L2	CO3
	b.	Find the support reactions and member forces for pin jointed plane truss shown in Fig Q6(b)	12	L3	CO3
 Fig. Q6(b)					
Module – 4					
Q.7	a.	Derive moment of Inertia of semicircle about its base and centroidal axis.	10	L2	CO4
	b.	Determine centroid of shaded portion with respect to given axis Fig Q7(b)	10	L3	CO3
 Fig Q7(b)					
OR					
Q.8	a.	Derive the centroid of a quadrant from first principle.	8	L2	CO4
	b.	Determine the centroid of shaded portion with respect to given axis. Ref Fig Q8(b)	12	L3	CO2
 Fig Q8(b)					

Module – 5					
Q.9	a.	Derive all the three basic equations of motion in Kinematics.	6	L2	CO4
	b.	Explain the following with the help of sketch for projectile motion. i) Horizontal range ii) Time of flight iii) Maximum height iv) Angle of projection.	6	L2	CO2
	c.	A body moves in a straight line as equation of motion is given by $S = 2t^3 - 4t + 10$. Determine : i) Time required for body to gain velocity 68 m/s from rest. ii) Acceleration of the body when velocity is 32m/s	8	L3	CO3
OR					
Q.10	a.	Defend the necessity of providing super elevation. How is it provided in the field?	5	L2	CO4
	b.	Define displacement, velocity, Distance travelled, speed and acceleration.	5	L2	CO2
	c.	A projectile is fired from the top of cliff 150m height with an initial velocity of 180m/s at an angle of inclination of 30° to horizontal. Determine: i) The greatest elevation above the cliff ii) The great elevation above the ground reached by particle iii) The horizontal distance from the point of projection to the point where projectile strikes the grounds.	10	L3	CO3

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Question Paper Version : A

First/Second Semester B.E./B.Tech./B.Ds. Degree Examination, Dec.2023/Jan.2024 Communicative English

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

-
1. 'Communication' The word originates from Laten word
 a) Communicm b) Communico c) Communicative d) None of these
 2. Communication is a _____ process.
 a) one way b) two - way c) three - way d) None of these
 3. In order to achieve the production target for a particular month, the discussion held between the production manager and supply manager will certainly be a perfect instance of _____ Communication.
 a) Vertical b) Spiral c) Cross wises d) Horizontal
 4. 'Poor Printing' it is a type of Barrier _____
 a) Verbal barrier b) Psychological barrier
 c) Physiological barrier d) Physical barrier
 5. 'Transcendental meditation' is example of _____ Communication.
 a) Extrapersonal b) Interpersonal c) Intrapersonal d) Non - verbal
 6. The most straight forward, basic interpersonal communication meaning is _____ Communication.
 a) Mass Communication b) Interpersonal
 c) Extrapersonal d) None of these
 7. Flashing eyes, avoiding eye contact may also cause _____ to effective Communication.
 a) Vernal barrier b) Listening barrier
 c) Psychological barrier d) Non – Verbal barrier
 8. In general oral Communication is the interchange of _____ between the sender and the receiver.
 a) cues and clues b) speaking c) gestures d) verbal messages

9. 'Follows a set pattern such as sequence of elements in a report' this is _____ Communication.
a) Professional b) General c) Personal d) None of these
10. In Organizations, informal communication also permeates. This informal communication is called _____.
a) Personal b) Professional c) Grapevine d) None of these
11. One native regional accent that has gained social prestige is the _____ of English.
a) Received pronunciation b) Reverse perception
c) Transcription d) None of these
12. A vowel sound whose quality does not change over the duration of the vowel is called _____.
a) Vowel b) Diphthong c) Pure vowel d) Consonant
13. The consonant sounds that come before vowel sound are called _____.
a) Coda b) Onset c) Medical d) None of these
14. What is the division of syllables of word 'Information'?
a) in-form-ation b) inform-ation c) in-formation d) in – for - ma - tion
15. Transcribe the word 'daughter'.
a) / 'dɔ : t ə / b) / dɔ 't ə / c) / dɔ : tr / d) / d 3 : t ə /
16. 'No. The women with the plastic bag'. In this sentence which words are having falling tone?
a) No, The woman b) Plastic bag c) No. bag d) None of these
17. Find the silent letter in the word 'though'.
a) th b) ou c) ugh d) gh
18. In the word 'Sheep' often mispronounced sound is
a) / ʃ / b) / S / c) ee d) p
19. Correct the spelling of the following word :
a) Renumeration b) Remuneration c) Remunation d) Remuration
20. How to pronounce the word 'Next'?
a) nekast b) nekest c) nekist d) nekst
21. I have _____ to buy a car. (Use appropriate noun)
a) no rupees b) no notes c) no money d) no moneys
22. _____ and _____ want to purchase a yacht. (fill with suitable pronoun).
a) He. I and you b) You, he and I c) I, he and you d) You, I and he
23. The flower smells _____. (Appropriate Adjective is)
a) sweet b) sweetly c) sweeter d) sweetest
24. It is _____ hotter today, (place an adverb).
a) most b) very c) more d) much

25. Hardly had he arrived _____ the portico crumbled. (Choose correct conjunction)
a) than b) then c) when d) None of these
26. We could barely do anything. (suitable question Tag).
a) Couldn't we? b) Could we? c) Can we? d) None of these
27. A Person who believes in God, we call him
a) Theist b) Atheist c) Eccentric d) Spinster
28. 'I saw a man' in this sentence which is pronounced with a weak verb, without any special emphasis.
a) I b) man c) saw d) a
29. Flour is made _____ wheat. (suitable preposition)
a) of b) by c) from d) in
30. _____ Humanity is in danger. (Choose the correct option).
a) A b) An c) The d) Not needed any article

Fill in the banks with appropriate prefixes :

31. The army people _____ attacked the enemy.
a) in - b) Inter c) Counter d) Un
32. My research Supervisor is _____ critical about everything.
a) hyper b) un c) mono d) intra

Choose the pairs of word / phrases :

33. Archive : Document
a) Warehouse : merchandise b) Library : Shelves
c) Theater : Plays d) Cinema : Projector
34. Coal : Mineral
a) Oxygen : Water b) river : dam c) gold : metal d) silver : mine

Choose the correct subject – verb concord in the sentences :

35. Diabetes _____ a silent killer.
a) is b) are c) were d) None of these
36. The Jury gave _____ verdict in an unbiased manner.
a) their b) them c) its d) None of these

Complete the sentence with appropriate tense from :

37. Sir, I _____ passed my B.Tech in 2003
a) have b) has c) was d) Not required
38. We _____ finished our assignment just now.
a) have b) has c) had d) None of these

Choose the correct option :

39. _____ was visible at this late hour.
a) No man b) No c) No men d) None of these

40. The tycoon had expired before the _____.
 a) Physician arrives b) Physician arriving
 c) Physician arrived d) Physician had arrived
41. When _____, I will tell him everything.
 a) he is coming b) he comes c) he will come d) None of the above
42. Which of these must be avoided in Presentation?
 a) Proper grammar b) Complex words c) Short sentence d) Clear voice
43. Which of these can be used to break the monotony in a speech?
 a) Humour b) Constant tone c) Low voice d) Sad story
44. Mother tongue influence can be effectively minimized in the classroom by _____.
 a) using the mother tongue more often b) giving examples from the mother tongue
 c) giving a lot of exposure in the target language
 d) giving inputs from the target language in a simple graded manner.
45. The art of clear and concise manner of speaking, with clarity of meaning and thought called _____.
 a) Elocution b) Latency c) Extempore d) Story telling

Read Comprehension : Read the passage carefully and select the suitable answer from the given options :

Quite recently India laid the foundation stone for one of its most sought – after projects running a Bullet Train. It was very well considered as a dream project of the Honorable Prime Minister. Entire India felt proud of having its first ever bullet train scheduled to run between Mumbai and Ahmedabad, a distance of 508 km in about 2 hours 35 minutes. In his own words, “To grow, one needs to expand one’s dreams and decide one’s strength to achieve that. It’s the New India which has to fly high”. “Bullet Train is a project that will provide pace to development. Along with new technology, it will also bring results faster”, he added. According to Achal Khare, the Managing Director of the National High speed Rail Corporation, the project would be completed by December 2005.

46. The _____ for the Bullet Train project was laid recently.
 a) railway track b) signaling system c) foundation stone d) None of these
47. The bullet train will take about _____ to run between both the States.
 a) 3 hrs, 25 min b) 2 hr, 55 min c) 2 hr, 35 min d) None of these
48. According to Prime Minister, the two benefits that the bullet train will bring are to provide pace to _____ to faster _____.
 a) travel , trains b) train, travel, results
 c) development, trains d) development, results
49. The bullet train schedule to run between _____ and _____.
 a) Mumbai and Ahmedabad b) Mumbai and Delhi
 c) Mumbai and Gujarat d) None of these
50. The project is expected to be _____ by December 2023.
 a) completed b) operationalised c) started d) None of these

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Question Paper Version : B

First/Second Semester B.E./B.Tech./B.Ds. Degree Examination, Dec.2023/Jan.2024
Communicative English

Time: 1 hr.]

[Max. Marks: 50

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Choose the correct option :

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Choose the pairs of word / phrases :

15. Archive : Document
a) Warehouse : merchandise b) Library : Shelves
c) Theater : Plays d) Cinema : Projector
16. Coal : Mineral
a) Oxygen : Water b) river : dam c) gold : metal d) silver : mine

Choose the correct subject – verb concord in the sentences :

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Complete the sentence with appropriate tense from :

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30. In Organizations, informal communication also permeates. This informal communication is called _____
a) Personal b) Professional c) Grapevine d) None of these
31. One native regional accent that has gained social prestige is the _____ of English.
a) Received pronunciation b) Reverse perception
c) Transcription d) None of these
32. A vowel sound whose quality does not change over the duration of the vowel is called _____
a) Vowel b) Diphthong c) Pure vowel d) Consonant
33. The consonant sounds that come before vowel sound are called _____
a) Coda b) Onset c) Medical d) None of these

34. What is the division of syllables of word 'Information'?
a) in-form-ation b) inform-ation c) in-for-ma-tion d) in - for - ma - tion
35. Transcribe the word 'daughter'.
a) / 'dɔ : t ə / b) / dɔ 't ə / c) / dɔ : tr / d) / d 3 : t ə /
36. 'No. The women with the plastic bag'. In this sentence which words are having falling tone?
a) No, The woman b) Plastic bag c) No. bag d) None of these
37. Find the silent letter in the word 'though'.
a) th b) ou c) ugh d) gh
38. In the word 'Sheep' often mispronounced sound is
a) / ʃ / b) / S / c) ee d) p
39. Correct the spelling of the following word :
a) Renumeration b) Remuneration c) Remunation d) Remuration
40. How to pronounce the word 'Next'?
a) nekast b) nekest c) nekist d) nekst
41. I have _____ to buy a car. (Use appropriate noun)
a) no rupees b) no notes c) no money d) no moneys
42. _____ and _____ want to purchase a yacht. (fill with suitable pronoun).
a) He. I and you b) You, he and I c) I, he and you d) You, I and he
43. The flower smells _____. (Appropriate Adjective is)
a) sweet b) sweetly c) sweeter d) sweetest
44. It is _____ hotter today, (place an adverb).
a) most b) very c) more d) much
45. Hardly had he arrived _____ the portico crumbled. (Choose correct conjunction)
a) than b) then c) when d) None of these
46. We could barely do anything. (suitable question Tag).
a) Couldn't we? b) Could we? c) Can we? d) None of these
47. A Person who believes in God, we call him
a) Theist b) Atheist c) Eccentric d) Spinster
48. 'I saw a man' in this sentence which is pronounced with a weak verb, without any special emphasis.
a) I b) man c) saw d) a
49. Flour is made _____ wheat. (suitable preposition)
a) of b) by c) from d) in
50. _____ Humanity is in danger. (Choose the correct option).
a) A b) An c) The d) Not needed any article

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Question Paper Version : C

First/Second Semester B.E./B.Tech./B.Ds. Degree Examination, Dec.2023/Jan.2024
Communicative English

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

Fill in the banks with appropriate prefixes :

1. The army people _____ attacked the enemy.
a) in - b) Inter c) Counter d) Un
2. My research Supervisor is _____critical about everything.
a) hyper b) un c) mono d) intra

Choose the pairs of word / phrases :

3. Archive : Document
a) Warehouse : merchandise b) Library : Shelves
c) Theater : Plays d) Cinema : Projector
4. Coal : Mineral
a) Oxygen : Water b) river : dam c) gold : metal d) silver : mine

Choose the correct subject – verb concord in the sentences :

5. Diabetes _____ a silent killer.
a) is b) are c) were d) None of these
6. The Jury gave _____ verdict in an unbiased manner.
a) their b) them c) its d) None of these

Complete the sentence with appropriate tense from :

7. Sir, I _____ passed my B.Tech in 2003
a) have b) has c) was d) Not required
8. We _____ finished our assignment just now.
a) have b) has c) had d) None of these

9. 'Communication' The word originates from Latin word
a) Communicm b) Communico c) Communicative d) None of these
10. Communication is a _____ process.
a) one way b) two - way c) three - way d) None of these
11. In order to achieve the production target for a particular month, the discussion held between the production manager and supply manager will certainly be a perfect instance of _____ Communication.
a) Vertical b) Spiral c) Cross wise d) Horizontal
12. 'Poor Printing' it is a type of Barrier _____
a) Verbal barrier b) Psychological barrier
c) Physiological barrier d) Physical barrier
13. 'Transcendental meditation' is example of _____ Communication.
a) Extrapersonal b) Interpersonal c) Intrapersonal d) Non - verbal
14. The most straight forward, basic interpersonal communication meaning is _____ Communication.
a) Mass Communication b) Interpersonal
c) Extrapersonal d) None of these
15. Flashing eyes, avoiding eye contact may also cause _____ to effective Communication.
a) Vernal barrier b) Listening barrier
c) Psychological barrier d) Non – Verbal barrier
16. In general oral Communication is the interchange of _____ between the sender and the receiver.
a) cues and clues b) speaking c) gestures d) verbal messages
17. 'Follows a set pattern such as sequence of elements in a report' this is _____ Communication.
a) Professional b) General c) Personal d) None of these
18. In Organizations, informal communication also permeates. This informal communication is called _____
a) Personal b) Professional c) Grapevine d) None of thee
19. One native regional accent that has gained social prestige is the _____ of English.
a) Received pronunciation b) Reverse perception
c) Transcription d) None of these
20. A vowel sound whose quality does not change over the duration of the vowel is called _____
a) Vowel b) Diphthong c) Pure vowel d) Consonant
21. The consonant sounds that come before vowel sound are called _____
a) Coda b) Onset c) Medical d) None of these
22. What is the division of syllables of word 'Information'?
a) in-form-ation b) inform-ation c) in-formation d) in - for - ma - tion

23. Transcribe the word 'daughter'.
 a) / 'dɔ : t ə / b) / dɔ 't ə / c) / dɔ : tr / d) / d 3 : t ə /
24. 'No. The women with the plastic bag'. In this sentence which words are having falling tone?
 a) No, The woman b) Plastic bag c) No. bag d) None of these
25. Find the silent letter in the word 'though'
 a) th b) ou c) ugh d) gh
26. In the word 'Sheep' often mispronounced sound is
 a) / ʃ / b) / S / c) ee d) p
27. Correct the spelling of the following word :
 a) Renumeration b) Remuneration c) Remunation d) Remuration
28. How to pronounce the word 'Next'?
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32. It is _____ hotter today, (place an adverb).
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33. Hardly had he arrived _____ the portico crumbled. (Choose correct conjunction)
 a) than b) then c) when d) None of these
34. We could barely do anything. (suitable question Tag).
 a) Couldn't we? b) Could we? c) Can we? d) None of these
35. A Person who believes in God, we call him
 a) Theist b) Atheist c) Eccentric d) Spinster
36. 'I saw a man' in this sentence which is pronounced with a weak verb, without any special emphasis.
 a) I b) man c) saw d) a
37. Flour is made _____ wheat. (suitable preposition)
 a) of b) by c) from d) in
38. _____ Humanity is in danger. (Choose the correct option).
 a) A b) An c) The d) Not needed any article

Read Comprehension : Read the passage carefully and select the suitable answer from the given options :

Quite recently India laid the foundation stone for one of its most sought – after projects running a Bullet Train. It was very well considered as a dream project of the Honorable Prime Minister. Entire India felt proud of having its first ever bullet train scheduled to run between Mumbai and Ahmedabad, a distance of 508 km in about 2 hours 35 minutes. In his own words, “To grow, one needs to expand one’s dreams and decide one’s strength to achieve that. It’s the New India which has to fly high”. “Bullet Train is a project that will provide pace to development. Along with new technology, it will also bring results faster”, he added. According to Achal Khare, the Managing Director of the National High speed Rail Corporation, the project would be completed by December 2005.

39. The _____ for the Bullet Train project was laid recently.
a) railway track b) signaling system c) foundation stone d) None of these
40. The bullet train will take about _____ to run between both the States.
a) 3 hrs, 25 min b) 2 hr, 55 min c) 2 hr, 35 min d) None of these
41. According to Prime Minister, the two benefits that the bullet train will bring are to provide pace to _____ to faster _____.
a) travel, trains b) train, travel, results
c) development, trains d) development, results
42. The bullet train schedule to run between _____ and _____.
a) Mumbai and Ahmedabad b) Mumbai and Delhi
c) Mumbai and Gujarat d) None of these
43. The project is expected to be _____ by December 2023.
a) completed b) operationalised c) started d) None of these

Choose the correct option :

44. _____ was visible at this late hour.
a) No man b) No c) No men d) None of these
45. The tycoon had expired before the _____.
a) Physician arrives b) Physician arriving
c) Physician arrived d) Physician had arrived
46. When _____, I will tell him everything.
a) he is coming b) he comes c) he will come d) None of the above
47. Which of these must be avoided in Presentation?
a) Proper grammar b) Complex words c) Short sentence d) Clear voice
48. Which of these can be used to break the monotony in a speech?
a) Humour b) Constant tone c) Low voice d) Sad story
49. Mother tongue influence can be effectively minimized in the classroom by _____.
a) using the mother tongue more often b) giving examples from the mother tongue
c) giving a lot of exposure in the target language
d) giving inputs from the target language in a simple graded manner.
50. The art of clear and concise manner of speaking, with clarity of meaning and thought called _____.
a) Elocution b) Latency c) Extempore d) Story telling

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Question Paper Version : D

First/Second Semester B.E./B.Tech./B.Ds. Degree Examination, Dec.2023/Jan.2024
Communicative English

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

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Choose the correct option :

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Choose the pairs of word / phrases :

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a) Warehouse : merchandise b) Library : Shelves
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Complete the sentence with appropriate tense from :

49. Sir, I _____ passed my B.Tech in 2003
a) have b) has c) was d) Not required
50. We _____ finished our assignment just now.
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* * * * *

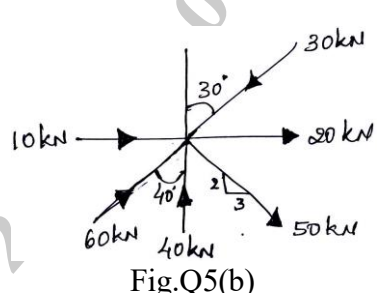
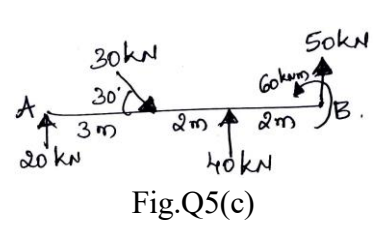
First Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan.2024

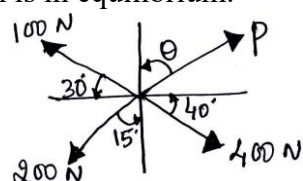
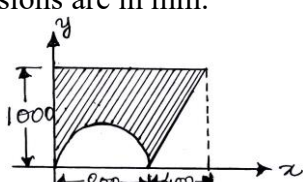
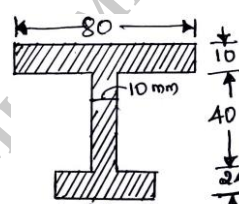
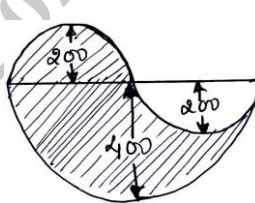
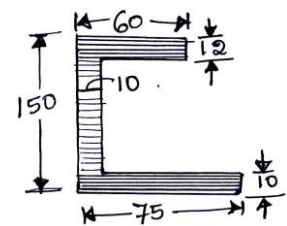
Introduction to Civil Engineering

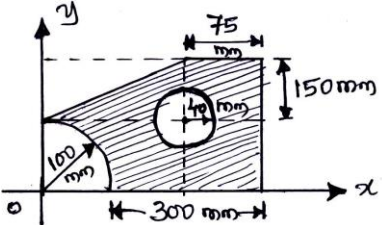
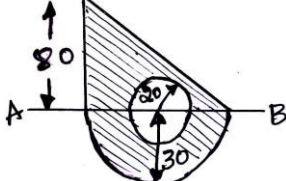
Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1				M	L	C
Q.1	a.	Explain briefly the scope of following specification of civil engineering: i) Environmental Engineering ii) Structural Engineering		08	L1	CO1
	b.	Write the composition of cement.		06	L1	CO1
	c.	What is brick? Write the classes of bricks.		06	L1	CO1
OR						
Q.2	a.	Differentiate Reinforced and Prestressed Concrete.		08	L1	CO1
	b.	Write a note on Construction Chemicals.		06	L1	CO1
	c.	Enumerate Structural Elements of a building.		06	L1	CO1
Module – 2						
Q.3	a.	Discuss on salient points on sustainable development goals.		08	L1	CO2
	b.	Write a note on Smart City Concept.		06	L1	CO2
	c.	Describe Solid Waste Management.		06	L1	CO2
OR						
Q.4	a.	Write the key points to be considered while identifying the landfills sites.		08	L1	CO2
	b.	Explain Refuse, Reuse and Recycle concepts.		06	L1	CO2
	c.	Write a note on Energy Efficient buildings.		06	L1	CO2
Module – 3						
Q.5	a.	Explain principles of superposition with a neat sketch.		04	L2	CO3
	b.	Determine the resultant of the system of forces shown in the Fig.Q5(b) below:  Fig.Q5(b)		08	L3	CO3
	c.	Determine the resultant of non concurrent system of forces shown in the Fig.Q5(c) below:  Fig.Q5(c)		08	L3	CO3
OR						
Q.6	a.	State and prove Varignon's theorem.		04	L2	CO3
	b.	Discuss resolution and composition of forces.		08	L2	CO3

	c.	Four forces of magnitude P, 100 N, 200 N, and 400 N are acting at a point as shown in Fig.Q6(c). Determine the magnitude and direction of force 'P' such that the force system is in equilibrium.	08	L3	CO3
 Fig.Q6(c)					
Module – 4					
Q.7	a.	Derive the centroid of a triangle having base 'b' and height 'h' from the first principles.	05	L2	CO4
	b.	Locate the centroid of the shaded area shown in Fig.Q7(b) with respect to reference axis. All dimensions are in mm.	08	L3	CO4
 Fig.Q7(b)					
	c.	Locate the centroid of the shaded area shown in Fig.Q7(c). All dimensions are in mm.	07	L3	CO4
 Fig.Q7(c)					
OR					
Q.8	a.	Derive the centroid of a rectangle of base 'b' and height 'h'.	05	L2	CO4
	b.	Determine the centroid of the shaded area shown in the Fig.Q8(b) below. All dimensions in mm.	08	L3	CO4
 Fig.Q8(b)					
	c.	Determine the centroid of the shaded area shown in the Fig.Q8(c). All dimensions in mm.	07	L3	CO4
 Fig.Q8(c)					

Module – 5					
Q.9	a.	State and prove parallel axis theorem.	05	L5	CO5
	b.	Derive the moment of inertia of a triangle of having base 'b' and height 'h' from the first principle.	05	L5	CO5
	c.	Find the polar moment of inertia of the plane lamina shown in Fig.Q9(c) about the point 'O' [I_{Oz}].	10	L5	CO5
 Note : Radius of circle is 40mm. Fig.Q9(c)					
OR					
Q.10	a.	Derive Moment of Inertia of a circle from the first principle.	05	L5	CO5
	b.	Define the terms i) Moment of Inertia ii) Radius of Gyration.	05	L5	CO5
	c.	Compute the MI of the area shown in Fig.Q10(c) about the axis AB. All dimensions are in mm only. Radius of circle is 20mm.	10	L5	CO5
 Fig.Q10(c)					

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First Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan.2024
Green Buildings

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	List and explain the uses of different types of cost effective Building Materials.	10	L2	CO1
	b.	Write a short note on Bamboo as a building material.	10	L2	CO1
OR					
Q.2	a.	What are the environmental issues related to quarrying of building materials.	10	L2	CO1
	b.	Write a short note on recycling of concrete as a building material.	10	L2	CO1
Module – 2					
Q.3	a.	Write a short note on cavity wall and Rat trap bond.	10	L2	CO2
	b.	Mention the advantages and disadvantages of pre engineered buildings.	10	L2	CO2
OR					
Q.4	a.	Briefly explain the contribution of Nirmiti Kendra for cost effective technology development.	10	L2	CO2
	b.	Write a short note on alternate roofing system → Filler slab and ferrocement.	10	L2	CO2
Module – 3					
Q.5	a.	Mention the causes and effects of Global warming.	10	L2	CO3
	b.	Explain the environmental benefits of Green Building.	10	L2	CO3
OR					
Q.6	a.	Mention the Benefits of energy efficient buildings.	10	L2	CO3
	b.	How buildings are contributing towards Global warming?	10	L2	CO3
Module – 4					
Q.7	a.	Briefly explain the BREEAM assessment category?	10	L2	CO4
	b.	Explain the criteria for a building rated as per GRIHA?	10	L2	CO4
1 of 2					

BETCK105B/BETCKB105					
OR					
Q.8	a.	List and explain the characteristics of sustainable building in Green Building design.	10	L2	CO4
	b.	List the objectives and principles of integrated life cycle design of materials and construction.	10	L2	CO4
Module – 5					
Q.9	a.	Explain the concept of solar passive heating in buildings.	10	L2	CO5
	b.	Define green composite. List the advantages and disadvantages of Green composites for building.	10	L2	CO5
OR					
Q.10	a.	Describe the process of management of solid waste.	10	L2	CO5
	b.	Explain the concept of Green cover and build environment.	10	L2	CO5

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Question Paper Version : A

First/Second Semester B.E. Degree Examination, Dec.2023/Jan.2024
Indian Constitution

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
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3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

-
1. The idea of a Constituent Assembly for India was put forwarded by
a) B.G. Tilak b) M. K. Gandhi c) M. N. Roy d) Motilal Nehru
 2. In Which year, did the Indian National Congress for the first time demand a Constituent Assembly?
a) 1934 b) 1935 c) 1936 d) 1937
 3. First time Constituent Assembly Constituted under the scheme formulated by the _____
a) Cabinet Mission Plan b) Mount Batten Plan
c) Non – Cooperation Movement d) None of these
 4. Members of the Constituent Assembly were _____
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50. _____ number of people represents graduate Constituency in the Vidhan Parishyadh
 - a) 1/10th
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Question Paper Version : B

First/Second Semester B.E. Degree Examination, Dec.2023/Jan.2024
Indian Constitution

Time: 1 hr.]

[Max. Marks: 50

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Question Paper Version : C

First/Second Semester B.E. Degree Examination, Dec.2023/Jan.2024
Indian Constitution

Time: 1 hr.]

[Max. Marks: 50

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Question Paper Version : D

First/Second Semester B.E. Degree Examination, Dec.2023/Jan.2024
Indian Constitution

Time: 1 hr.]

[Max. Marks: 50

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28. Who summons, prorogues and dissolves the State Legislative?
 - a) Chief Minister
 - b) Speaker
 - c) Governor of the State
 - d) President of India
29. Minimum and Maximum strength of the Legislative Assembly is
 - a) 60 and 500
 - b) 40 and 100
 - c) 70 and 700
 - d) 100 and 1000
30. _____ number of people represents graduate Constituency in the Vidhan Parishyadh
 - a) 1/10th
 - b) 2/3rd
 - c) 1/2th
 - d) 1/5th
31. It is a law passed to give effect to directive principles of State Policy.
 - a) Equal Pay for Equal Work Act
 - b) Right to Information Act
 - c) Juvenile Delinquency Act
 - d) Passport Act
32. Fundamental Duties are _____
 - a) Enforceable
 - b) Absolute
 - c) Restricted
 - d) Non - enforceable
33. Free Legal Aid is applicable to
 - a) APL Families
 - b) Financial Incapable Persons
 - c) NRI Families
 - d) CPI Families
34. It shall be duty of the every Citizen of India
 - a) Value our Culture
 - b) Renounce Foreign Culture
 - c) Impose our Culture
 - d) Preserve Western Culture
35. Fundamental Rights and Directive Principles are
 - a) Complementary to each other
 - b) Opposite to each other
 - c) Extension to each other
 - d) None of these
36. All the Executive powers of Union shall vest with
 - a) The Prime Minister
 - b) The President
 - c) The Parliament
 - d) The People

37. Who is Competent Authority in India to declare War?
a) Defense Minister b) Prime Minister c) Chief of the Army d) President
38. Union Legislature shall be _____
a) Bi Cameral b) Uni Cameral c) Tri Cameral d) Any Cameral
39. There shall not be more than _____ gap between two sessions of the Parliament.
a) 6 months b) 6 weeks c) 6 days d) 6 years
40. The Prime Minister is _____
a) Elected b) Appointed c) Nominated d) Both (a) & (b)
41. India can make its own laws because, India is
a) Independent b) Sovereign c) Secular d) Democratic
42. Preamble is
a) Part of the Constitution b) Amendable part
c) Non – Amendable d) Both ‘a’ and ‘c’
43. Fundamental Rights are _____
a) Enforceable b) Absolute c) Extraterritorial d) None of these
44. “Rights are not only the privileges, they are the weapons in hands of citizens to control the Arbitrary Government” – who gave this statement?
a) Dr. Baby Rajendra Prasad b) S.V. Patel
c) J. L. Nehru d) Dr. B. R. Ambedkar
45. The Preamble to the Constitution declares India as _____
a) Sovereign , Democratic , Republic
b) Sovereign, Socialist, Secular, Democratic, Republic
c) Socialist , Democratic , Republic
d) Sovereign , Democratic , Republic , Secular.
46. The Preamble aims to secure _____
a) Fundamental Rights b) Fundamental Duties
c) Dignity of the Individual d) Security of the People
47. Freedom to form an Assembly can be restricted on the ground
a) Public order b) Morality c) Health d) Wealth
48. Fundamental Rights are not applicable to
a) Foreigner b) Criminals
c) Members of Armed Forces d) Both ‘a’ & ‘c’
49. For enforcement of Fundamental Rights Supreme Court can issue
a) An order b) Direction c) Writs d) All of these
50. Provisions of Art. 20 applicable only in
a) Criminal cases b) Civil cases c) Administrative cases d) Military cases

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BMATC101

First Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan.2024 Mathematics – I for Civil Engineering Stream

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.
3. VTU formula handbook is permitted.*

Module – 1			M	L	C
Q.1	a.	With usual notations prove that $\frac{1}{p^2} = \frac{1}{r^2} + \frac{1}{r^4} \left(\frac{dr}{d\theta} \right)^2.$	6	L2	CO1
	b.	Find the angle between the radius vector and tangent to the curve $r = a(1+\cos \theta)$ at $\theta = \frac{\pi}{3}$.	7	L2	CO1
	c.	Find the radius of curvature of the curve $y = 4 \sin x - \sin 2x$ at $x = \frac{\pi}{2}$.	7	L2	CO1
OR					
Q.2	a.	Find the pedal equation of the curve $r^n \cos n \theta = a^n$.	8	L2	CO1
	b.	Find the radius of curvature of the curve $r = a(1-\cos \theta)$.	7	L2	CO1
	c.	Using modern mathematical tool write a programme to plot the curve $r = 2(\cos 2 \theta)$.	5	L2	CO1
Module – 2					
Q.3	a.	Expand $\sqrt{1 + \sin 2x}$ using Maclaurin's series upto the term containing x^4 .	6	L2	CO2
	b.	If $Z = f(x + ct) + g(x - ct)$ prove that $\frac{\partial^2 Z}{\partial t^2} = c^2 \frac{\partial^2 Z}{\partial x^2}$.	7	L2	CO2
	c.	Show that $f(x, y) = x^3 + y^3 - 3x - 12y + 20$ has a maximum value at the point $(-1, -2)$ and minimum value at the point $(1, 2)$.	7	L3	CO2
OR					
Q.4	a.	If $U = f(2x - 3y, 3y - 4z, 4z - 2x)$ then prove that $\frac{1}{2} \frac{\partial u}{\partial x} + \frac{1}{3} \frac{\partial u}{\partial y} + \frac{1}{4} \frac{\partial u}{\partial z} = 0.$	8	L2	CO2
	b.	If $u = x^2 + y^2 + z^2, v = xy + yz + zx, w = x + y + z$, Find $\frac{\partial(u, v, w)}{\partial(x, y, z)}$.	7	L2	CO2

	c.	Using modern mathematical tool, write a programme to evaluate $\lim_{x \rightarrow \infty} (1 + \frac{1}{x})^{\frac{1}{x}}$.	5	L3	CO5
Module – 3					
Q.5	a.	Solve : $\tan y \frac{dy}{dx} + \tan x = \cos y \cos^2 x$.	6	L2	CO3
	b.	Find the orthogonal trajectories of the family of curves $r = 2a \cos \theta$ where 'a' is a parameter.	7	L3	CO3
	c.	Solve $p^2 + p(x + y) + xy = 0$.	7	L2	CO3
OR					
Q.6	a.	Solve $(x^2 + y^3 + bx) dx + xy^2 dy = 0$.	6	L2	CO3
	b.	Water at temperature 10°C takes 5 minutes to warm upto 20°C in a room temperature of 40°C . Find the temperature of the water after 20 minutes.	7	L2	CO3
	c.	Solve $(px - y)(py + x) = 2p$ by reducing into Clairaut's form taking substitution $X = x^2$ and $Y = y^2$.	7	L3	CO3
Module – 4					
Q.7	a.	Solve $\frac{d^3y}{dx^3} + \frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 0$.	6	L2	CO3
	b.	Solve $(D - 2)^2 y = 2(e^{2x} + \sin 2x)$.	7	L2	CO3
	c.	Solve $x^2 \frac{d^2y}{dx^2} - 5x \frac{dy}{dx} + 8y = 2 \log x$.	7	L2	CO3
OR					
Q.8	a.	Solve $\frac{d^2y}{dx^2} + y = x^2 + 2x + 4$.	6	L2	CO3
		Using method of variation of parameters Solve $\frac{d^2y}{dx^2} + y = \sec x$	7	L2	CO3
	c.	Solve $(1+2x)^2 \frac{d^2y}{dx^2} - 6(1+2x) \frac{dy}{dx} + 16y = 8(1+2)x$.	7	L2	CO3
Module – 5					
Q.9	a.	Find the rank of the matrix : $\begin{bmatrix} 21 & 22 & 23 & 24 \\ 22 & 23 & 24 & 25 \\ 23 & 24 & 25 & 26 \\ 24 & 25 & 26 & 27 \end{bmatrix}$	6	L2	CO4

	b.	Test for consistency and solve the following system of equations : $x + 2y + 3z = 1$, $2x + 3y + 8z = 2$, $x + y + z = 3$.	7	L2	CO4
	c.	Solve the following system of equations using Gauss Jordan method : $x + y + z = 9$, $2x + y - z = 0$, $2x + 5y + 7z = 52$.	7	L3	CO4
OR					
Q.10	a.	Find the rank of the matrix $\begin{bmatrix} -2 & -1 & -3 & -1 \\ 1 & 2 & 3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & -1 & -1 \end{bmatrix}$	6	L2	CO4
	b.	Use Gauss – Seidel Iteration method to solve the system of equations $5x + 2y + z = 12$, $x + 4y + 2z = 15$, $x + 2y + 5z = 20$ (upto 3 iterations).	7	L3	CO4
	c.	Using modern mathematical tool, write a programme to find the largest eigen vector if $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$	7	L3	CO5

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BPHYC102/202

First/Second Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan.2024 Applied Physics for Civil Engineering Stream

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. VTU Formula Hand Book is permitted.
3. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	What is Force Constant? Obtain an expression for effective spring constant of two springs connected in series and parallel.	8	L2	CO1
	b.	Describe the construction and working of Hand operated Reddy shock tube with neat diagram.	7	L2	CO1
	c.	The displacement of two springs connected in series and parallel is respectively 3.86×10^{-2} m and 0.8×10^{-2} m , for a mass 0.05kg. Calculate the force constant of springs when connected in series and parallel. [given $g = 9.8 \text{ m/s}^2$].	5	L3	CO5
OR					
Q.2	a.	Mentioning the various forces acting on a system under damped oscillations, setup differential equation for damped oscillations and discuss in brief the types of damping with graphical representation.	7	L3	CO1
	b.	Define the terms Mach numbers and Mach angle. List the characteristics and applications of shock waves (any 3).	8	L2	CO1
	c.	In a Reddy shock tube experiment, the time taken to travel between two pressure sensors is 250 μs . If the distance between the sensors is 120mm. Calculate Mach number. [Given speed of sound 340 m/s].	5	L3	CO1
Module – 2					
Q.3	a.	State Hooke's Law. Explain stress hardening and softening with the help of stress – strain diagram.	7	L2	CO1
	b.	Define Bending Moment. Derive an expression for bending moment with help of a sketch.	9	L3	CO1
	c.	Calculate the extension produced in a wire of length 2m and radius 0.013×10^{-2} m due to a force of 14.7 N, applied along its length. Given Young's modulus of material of wire $Y = 2.1 \times 10^{11} \text{ N/m}^2$.	4	L3	CO1
OR					
Q.4	a.	Define Young's modulus. Rigidity modulus and Poisson's ratio. Derive the relation between them.	10	L2	CO1
	b.	What is Fracture? Discuss the failure of Engineering materials.	6	L2	CO1

	c.	A solid lead sphere of radius 10.3m is subjected to a normal pressure of 10N/m^2 acting all over the surface. Determine change in its volume. Given $K = 4.58 \times 10^{10} \text{ N/m}^2$.	4	L3	CO1
Module – 3					
Q.5	a.	Explain the requisites for good acoustics.	5	L2	CO2
	b.	Define Reverberation time and hence derive Sabine's formula.	10	L3	CO2
	c.	A Cinema hall has a volume of 7500m^3 with reverberation time of 1.5sec. What should be the total absorption in the hall?	5	L2	CO2
OR					
Q.6	a.	Explain the impact of noise in multi – stored buildings.	6	L2	CO2
	b.	What is the difference between Radiometry and Photometry? Define the terms Radiant Energy, Radiant Power , Radiant Intensity and Irradiance with equations.	10	L2	CO2
	c.	A classroom has dimensions $20 \times 15 \times 5 \text{ m}^3$. The reverberation time is 3.5sec. Calculate the total absorption of its surfaces and the average absorption coefficient.	4	L3	CO2
Module – 4					
Q.7	a.	Define Population Inversion. With a neat labeled diagram, describe the construction and working of semiconductor laser.	8	L2	CO3
	b.	Define Acceptance Angle. Derive an expression for numerical aperture of an optical fiber.	8	L2	CO3
	c.	In a diffraction grating experiment, the light undergoes 2 nd order diffraction at an angle of 22° . Find the wavelength of laser source given the slit width is $1.66 \times 10^{-4} \text{ cm}$.	4	L3	CO5
OR					
Q.8	a.	Discuss the principle, construction and working of optical fiber displacement sensor.	7	L2	CO1
	b.	Mention the requisites of laser system. Write a note on application of lasers in defence (Laser Range finder) and in the measurement of atmospheric pollutants (LIDAR).	9	L2	CO1
	c.	An optical signal losses 15% of its power after travelling a fiber length 400. What is the fiber loss?	4	L2	CO1
Module – 5					
Q.9	a.	What is an Earthquake? How do we classify earthquakes and explain?	9	L2	CO4
	b.	Discuss how Tsunami waves are created and what are their adverse effects?	7	L2	CO4

	c.	Calculate the intensity of an earthquake of magnitude 7.1 assuming the base intensity as I_0 .	4	L3	CO4
OR					
Q.10	a.	Explain how excess rainfall induces landslides.	7	L2	CO4
	b.	Name any two fire proofing materials. List fire safety and prevention measures required at your workplace.	8	L2	CO4
	c.	Intensity of one earthquake is 40 times the intensity of the other. If the magnitude of first earthquake is 8.5, estimate the magnitude of the other.	5	L3	CO4

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